Ex. N0:3

Depth First Search

AIM:

To write a program to solve any problem using Depth First Search in SWI Prolog.

ALGORITHM:

Step 1: Start from the given starting node.

Step 2: Mark the current node as visited.

Step 3: Check whether the current node is the goal.

Step 4: If it is the goal, return the path.

Step 5: Otherwise, find an unvisited node connected to the current node.

Step 6: Move to that node.

Step 7: Mark the new node as visited.

Step 8: Repeat Steps 3–7 recursively.

Step 9: If there is no unvisited connected node, backtrack to the previous node.

Step 10: Continue until the goal is found or there are no more nodes to explore.

PROGRAM:

```
edge(a, b).
edge(a, c).
edge(b, d).
edge(b, e).
edge(c, f).
edge(c, g).
edge(e, h).
edge(f, i).

dfs(Start, Goal, Path) :-
   dfs_search(Start, Goal, [Start], Path).

dfs_search(Goal, Goal, Visited, Path) :-
   reverse(Visited, Path).

dfs_search(Current, Goal, Visited, Path) :-
   edge(Current, Next),
   \+ member(Next, Visited),
   dfs_search(Next, Goal, [Next|Visited], Path).
```

OUTPUT:



RESULT:

Thus the Depth First Search program was executed successfully in SWI Prolog console and verified.